or $y\dfrac{1}{x\cos x} = x + C$

or $y = x^2\cos x + Cx\cos x$ is the required solution.

Example 10.24

Solve : $\dfrac{dy}{dx} + 2y\cot x = 3x^2\operatorname{cosec}^2 x$.

Solution

Given that the equation is $\dfrac{dy}{dx} + 2y\cot x = 3x^2\operatorname{cosec}^2 x$.

This is a linear differential equation. Here, $P = 2\cot x$; $Q = 3x^2\operatorname{cosec}^2 x$.

$$\int P dx = \int 2\cot x dx = 2\log|\sin x| = \log|\sin x|^2 = \log\sin^2 x.$$

$$\text{Thus, I.F} = e^{\int P dx} = e^{\log\sin^2 x} = \sin^2 x.$$

$$\text{Hence, the solution is. } ye^{\int P dx} = \int Qe^{\int P dx} dx + C.$$

$$\text{That is, } y\sin^2 x = \int 3x^2\operatorname{cosec}^2 x\cdot\sin^2 x dx + C = \int 3x^2 dx + C = x^3 + C.$$

Hence, $y\sin^2 x = x^3 + C$ is the required solution.

Example 10.25

Solve $\left(1+x^3\right)\dfrac{dy}{dx} + 6x^2 y = 1 + x^2$.

Solution

Here, to make the coefficient of $\dfrac{dy}{dx}$ unity, divide both sides by $\left(1+x^3\right)$.

Then the equation is $\dfrac{dy}{dx} + \dfrac{6x^2 y}{1+x^3} = \dfrac{1+x^2}{1+x^3}$.

This is a linear differential equation in y.

$$\text{Here, } P = \frac{6x^2}{1+x^3}; Q = \frac{1+x^2}{1+x^3}$$

$$\int P dx = \int \frac{6x^2}{1+x^3} dx = 2\log\left|1+x^3\right| = \log\left|1+x^3\right|^2 = \log\left(1+x^3\right)^2$$

$$\text{Thus, I.F.} = e^{\int P dx} = e^{\log\left(1+x^3\right)^2} = \left(1+x^3\right)^2$$

$$\text{Hence the solution is } ye^{\int P dx} = \int Qe^{\int P dx} dx + C.$$

$$\text{That is, } y\left(1+x^3\right)^2 = \int \frac{1+x^2}{1+x^3}\left(1+x^3\right)^2 dx + C = \int\left(1+x^2\right)\left(1+x^3\right) dx + C = \int\left(1+x^2+x^3+x^5\right) dx + C$$

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